

Use Case Diagram and UML Diagrams

ShieldAPI

Prepared by:

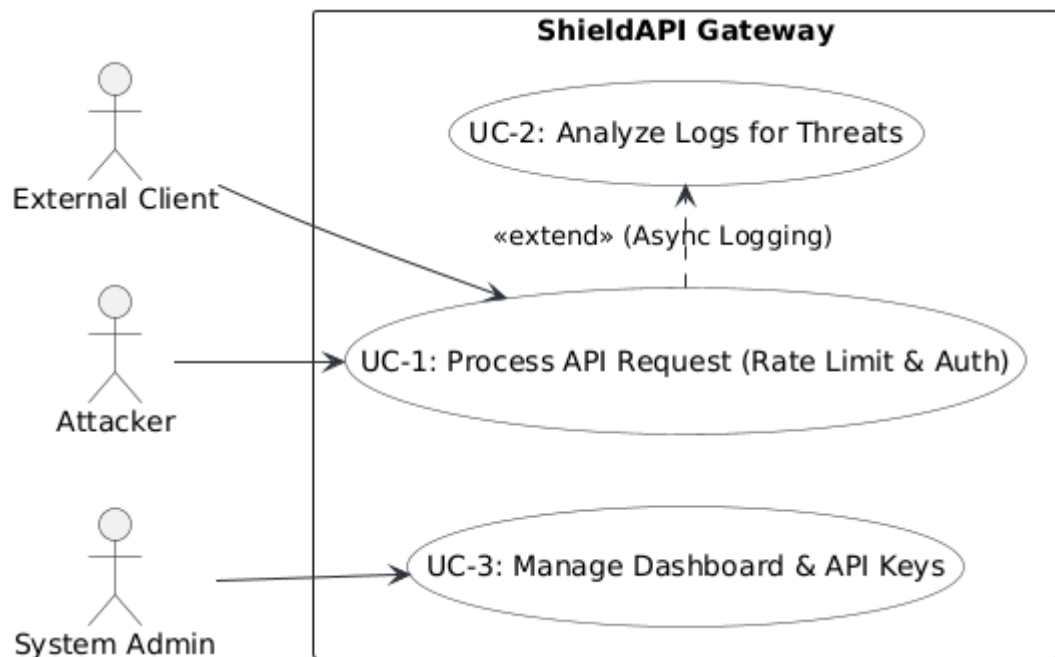
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1. Use Case Diagram



Brief Use Case Descriptions

UC-1: Process API Request (Rate Limit & Auth)

- **Purpose:** Validates API keys, enforces token-bucket rate limits, and routes safe traffic to the backend.
- **Actor(s) Involved:** External Client, Attacker.
- **Expected Outcome:** Safe requests successfully reach the backend; invalid or excessive requests are immediately rejected with 401 or 429 error codes.

UC-2: Analyze Logs for Threats

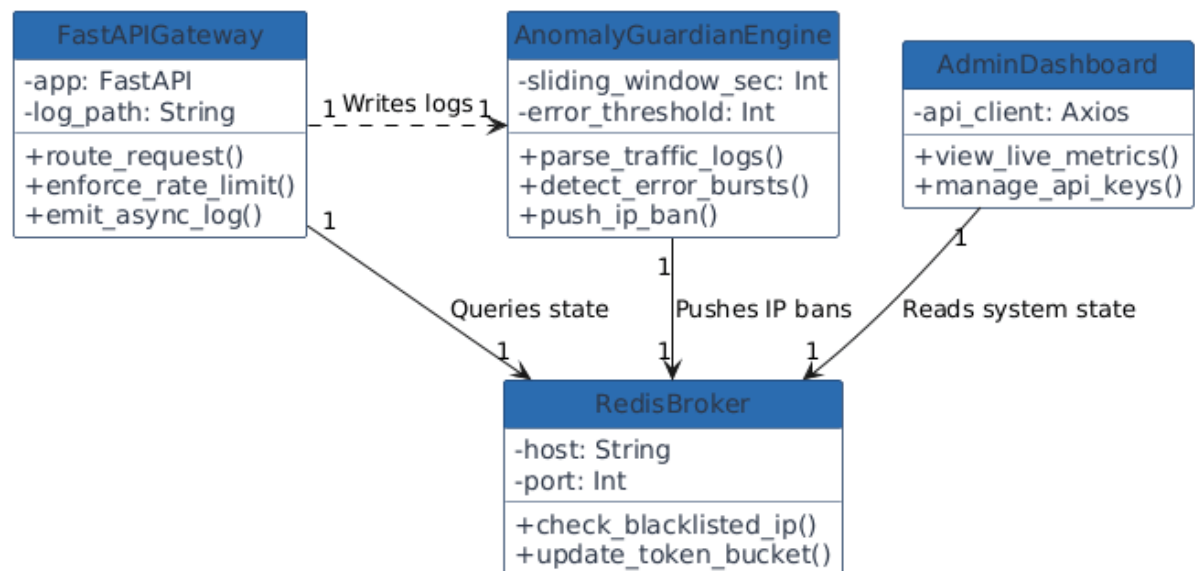
- **Purpose:** Asynchronously reads access logs to detect volumetric brute-force attacks without slowing down the main gateway.
- **Actor(s) Involved:** ShieldAPI Gateway (via Async Logging).
- **Expected Outcome:** When an error burst breaches the heuristic threshold, malicious IPs are automatically identified and pushed to the Redis blacklist.

UC-3: Manage Dashboard & API Keys

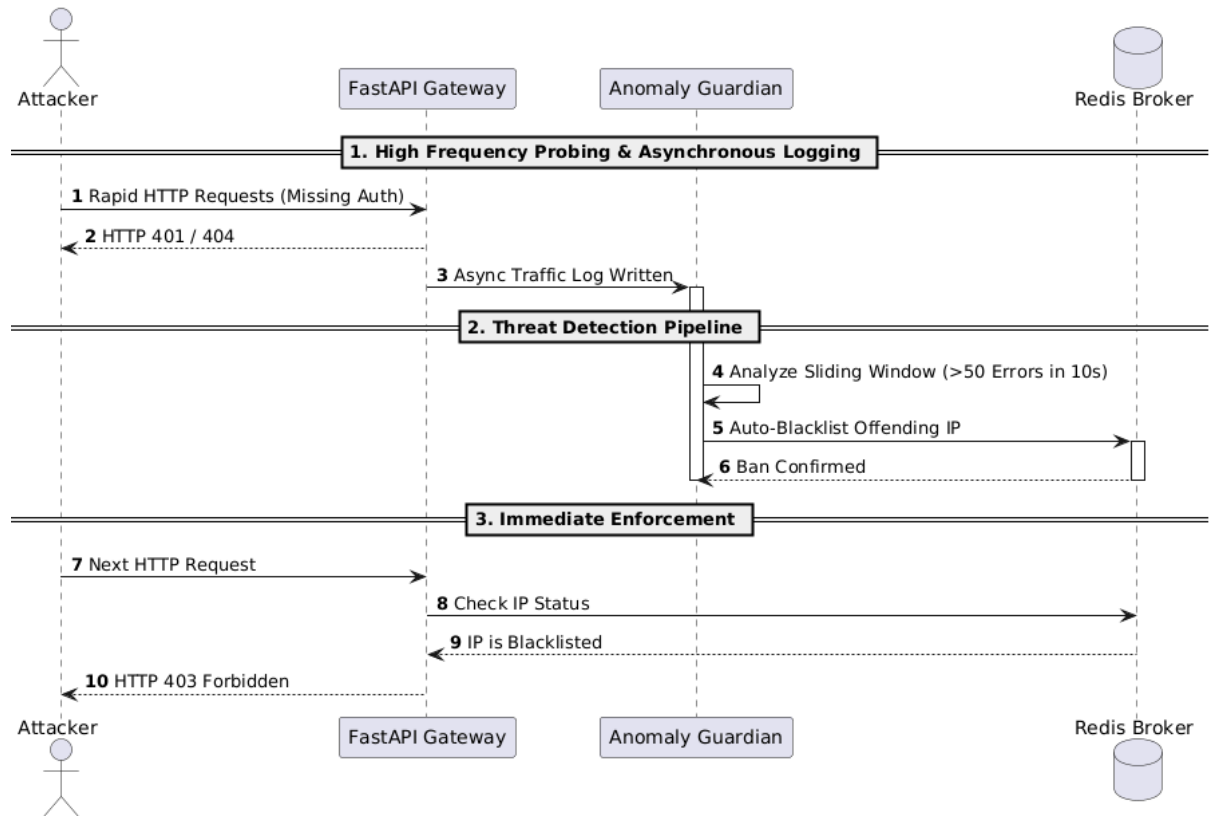
- **Purpose:** Allows the administrator to view real-time system health, telemetry, and manage access credentials via a React interface.
- **Actor(s) Involved:** System Admin.
- **Expected Outcome:** The administrator can successfully monitor live traffic and actively create or revoke API keys stored in the broker.

2. UML Diagrams

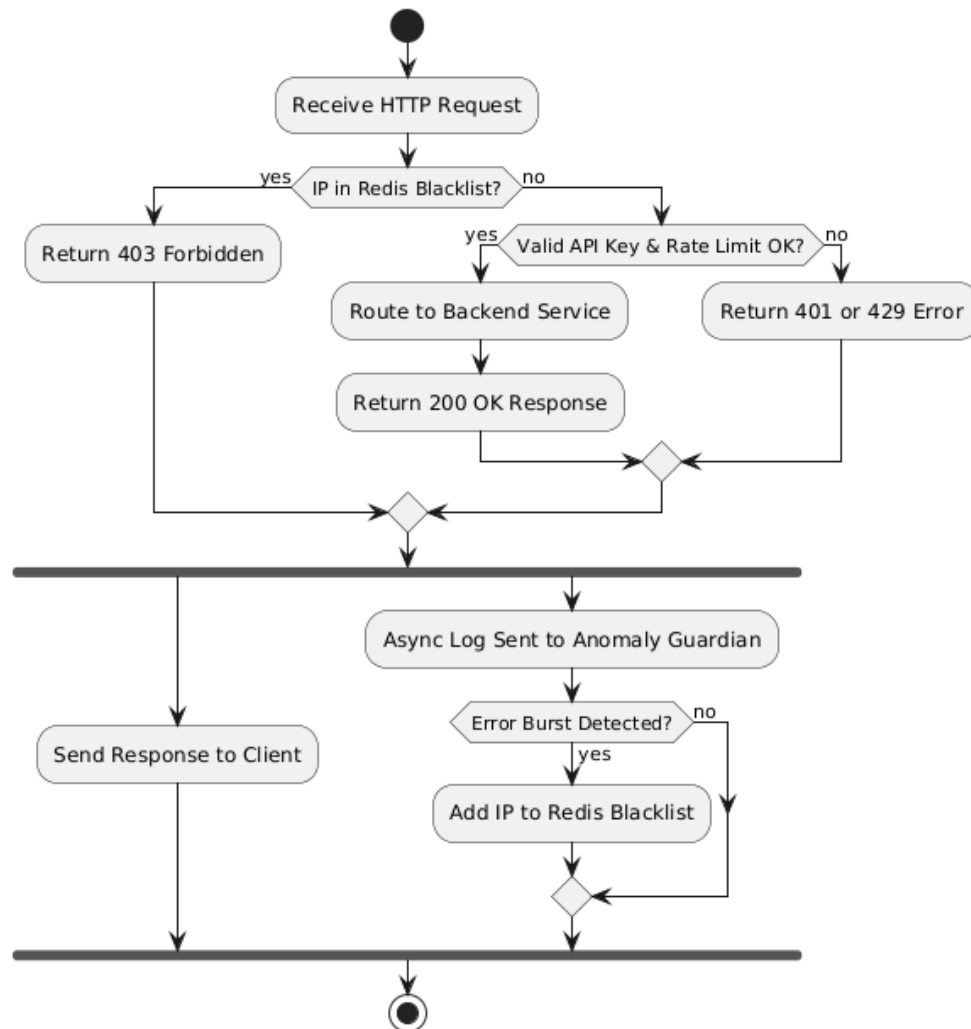
a) Class Diagram:



b) Sequence Diagram



c) Activity Diagram



d) State Diagram

